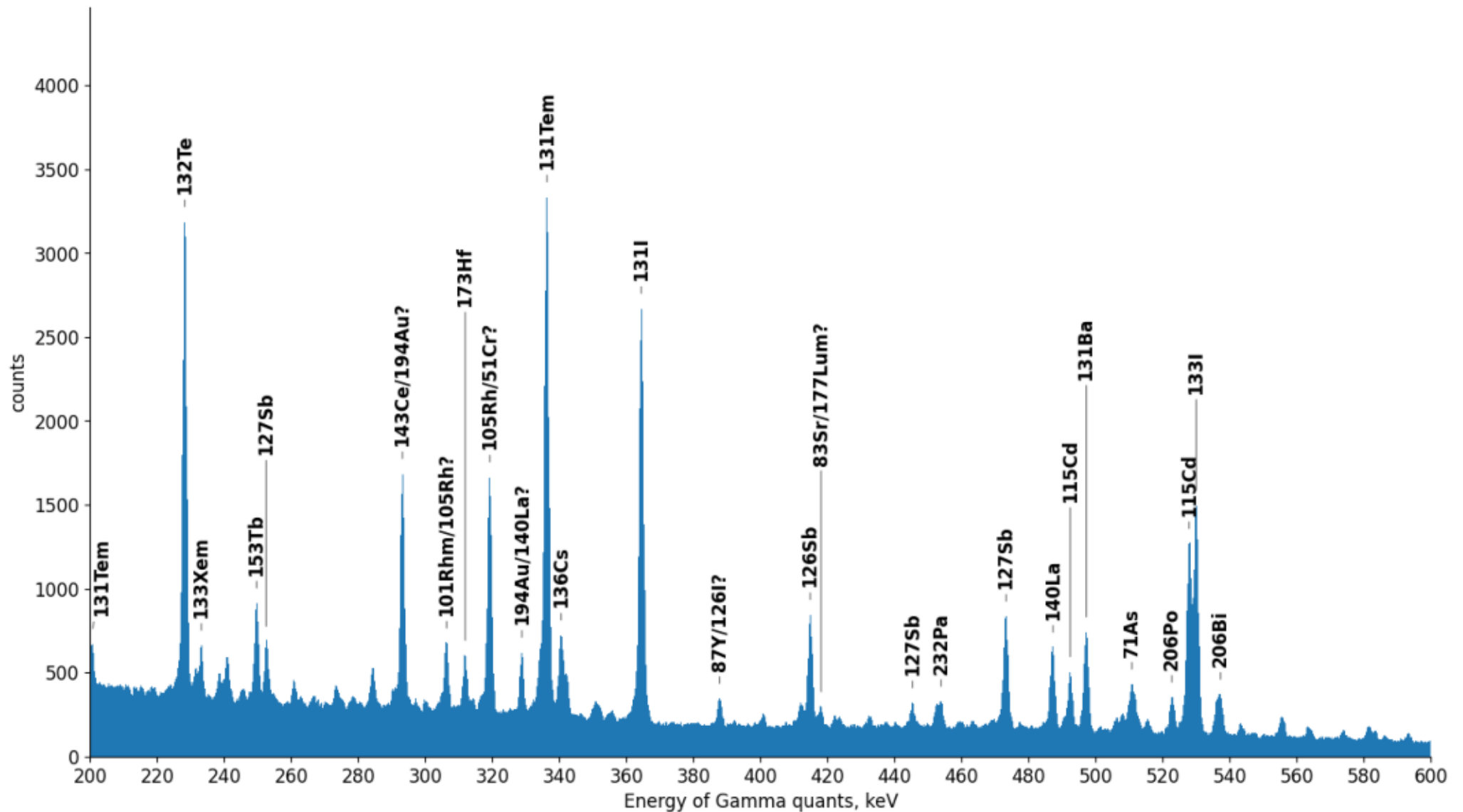


API-based tools — sample outputs

The gamma-spectrum analyzer and the nuclide chart viewer running on the public IAEA Live Chart of Nuclides API (nds.iaea.org) instead of the curated gamma library and the IsotopeDB MySQL database.

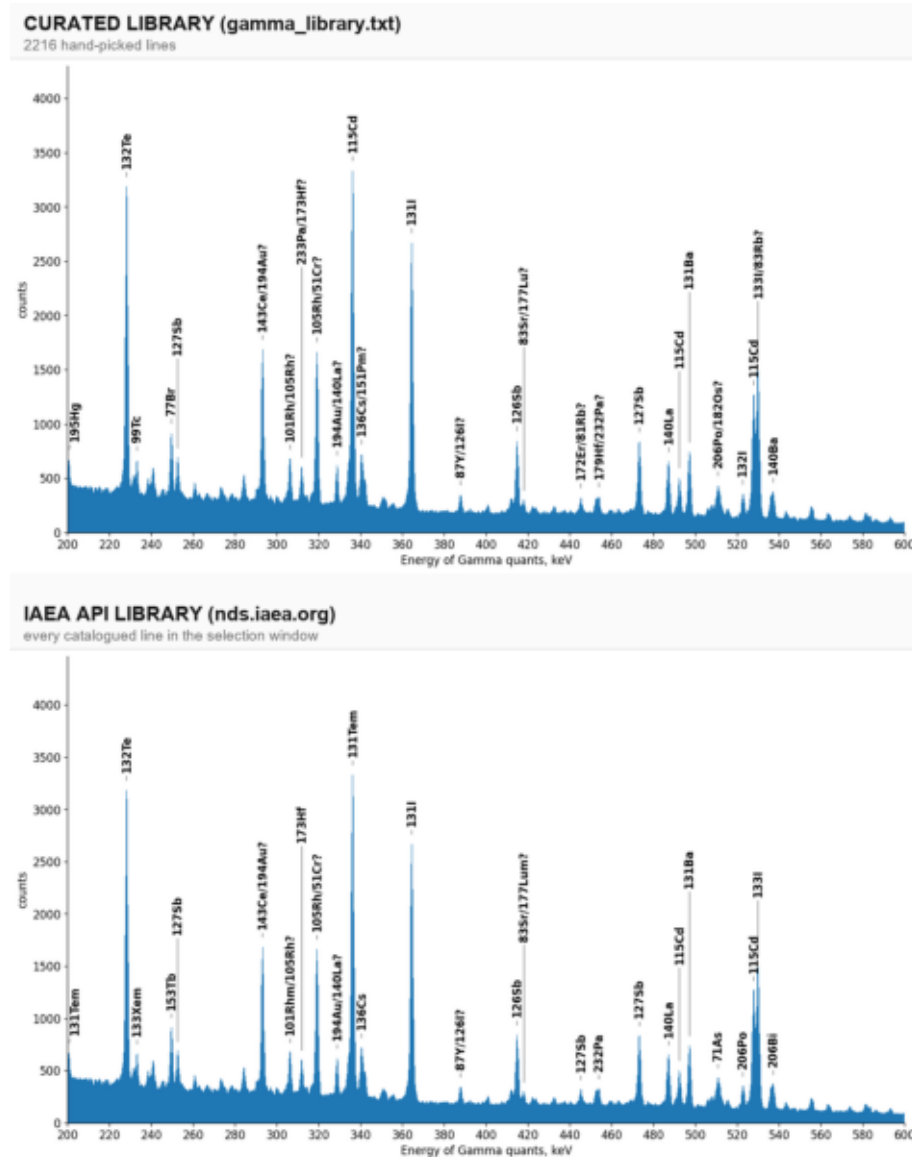
Spectrum figures: the bundled test_spectrum.Spe (1083 s acquisition, 5.04 days after a 119-minute proton irradiation), identified against the 50 538-line library assembled from per-nuclide decay radiation data. Chart figures: the isotope ^{81}Ga from the bulk ground-states table (3386 nuclides, one request).

Summary spectrum with IAEA API identifications, 200-600 keV



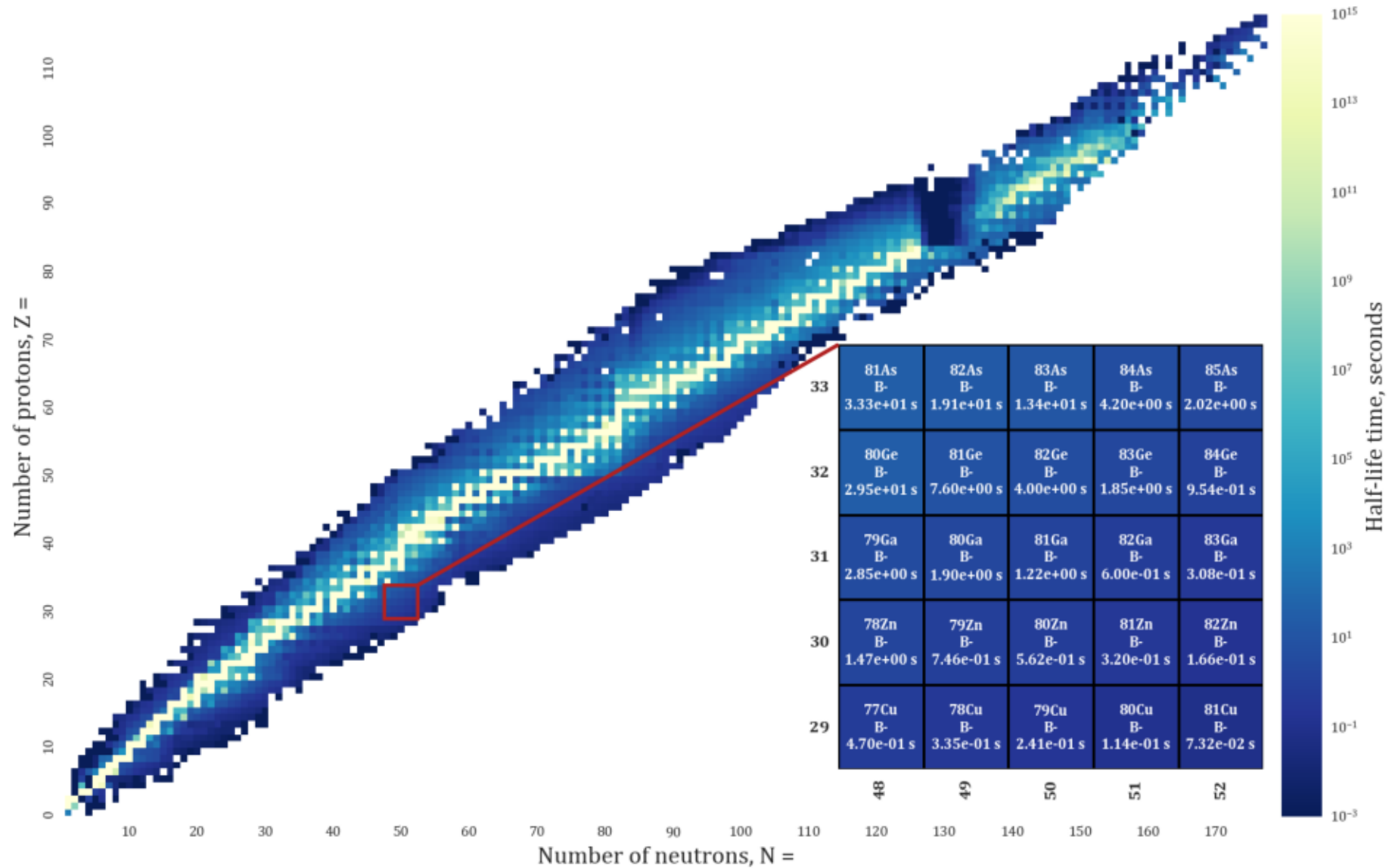
The same 59 detected peaks as the file-based analyzer -- every fit and activity number is identical; only the identification source differs. Metastable parent states are named explicitly (131Te, 101Rh) because the API reports the emitting state.

Curated library vs IAEA API library



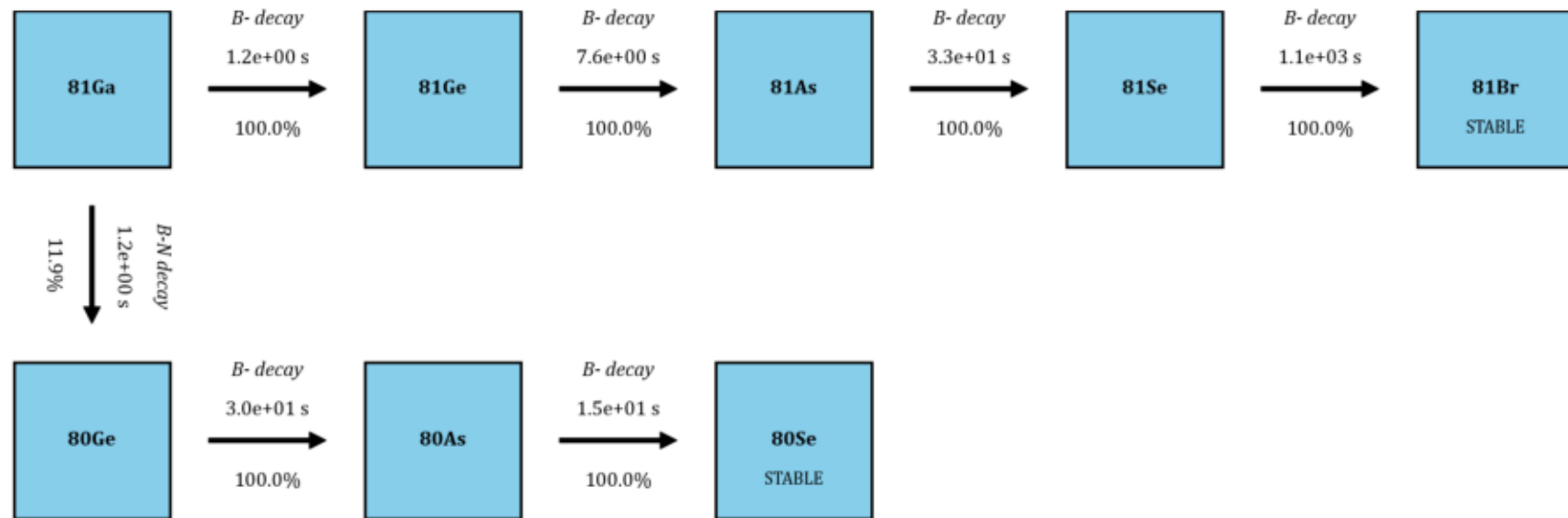
Identifications agree for 43 of 59 peaks. The differences trace to three causes. (1) The curated library encodes transient equilibrium by listing generator-fed daughters under the rate-limiting parent half-life -- its 140.5 keV entry carries the 66 h of 99Mo rather than the 6 h of 99mTc, so it identifies the line days after irradiation where the raw state half-life ranks it away. (2) Natural background lines such as 40K at 1461 keV score poorly in a production-and-decay ranking. (3) The API library offers ~25x more candidate lines within the match window.

Chart of nuclides from the ground-states table



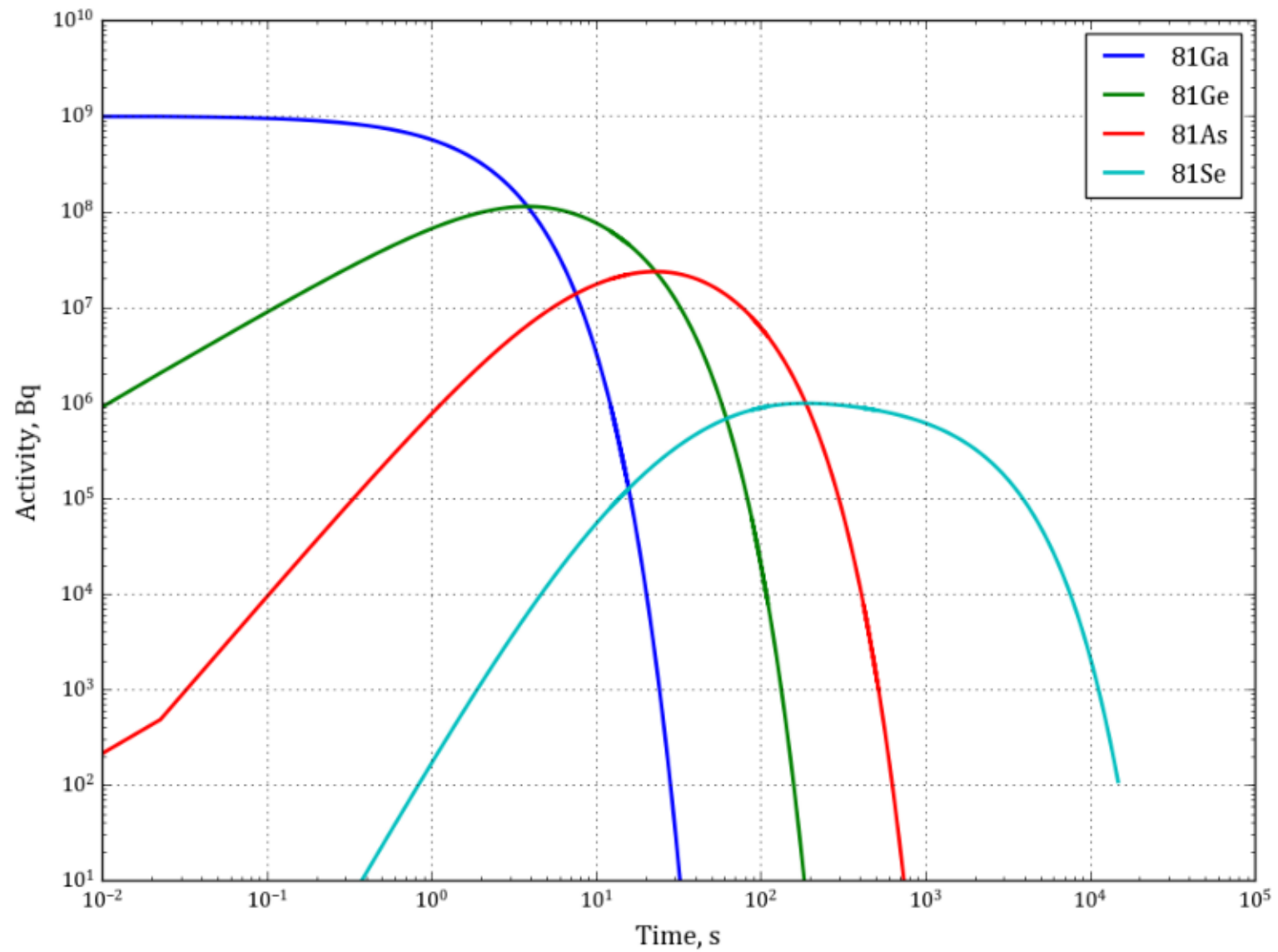
Half-life heatmap over all 3386 IAEA nuclides with the inset on 81Ga. Rendered by the same figure code as the database version; against the local data, half-lives agree within 5% for 90% of the 2820 common unstable nuclides, and several placeholder values from the 2013 NuDat export are replaced by measured ones.

Decay chain of 81Ga from API data



Identical member for member to the database version: the main chain $^{81}\text{Ga} > ^{81}\text{Ge} > ^{81}\text{As} > ^{81}\text{Se}$ to stable ^{81}Br , and the 11.9% beta-delayed-neutron branch to $^{80}\text{Ge} > ^{80}\text{As} > ^{80}\text{Se}$.

Activity of the chain members



Bateman activity from 1 GBq of ^{81}Ga , computed by matrix exponential of the decay matrix built from API branching ratios and half-lives. Indistinguishable from the database version.